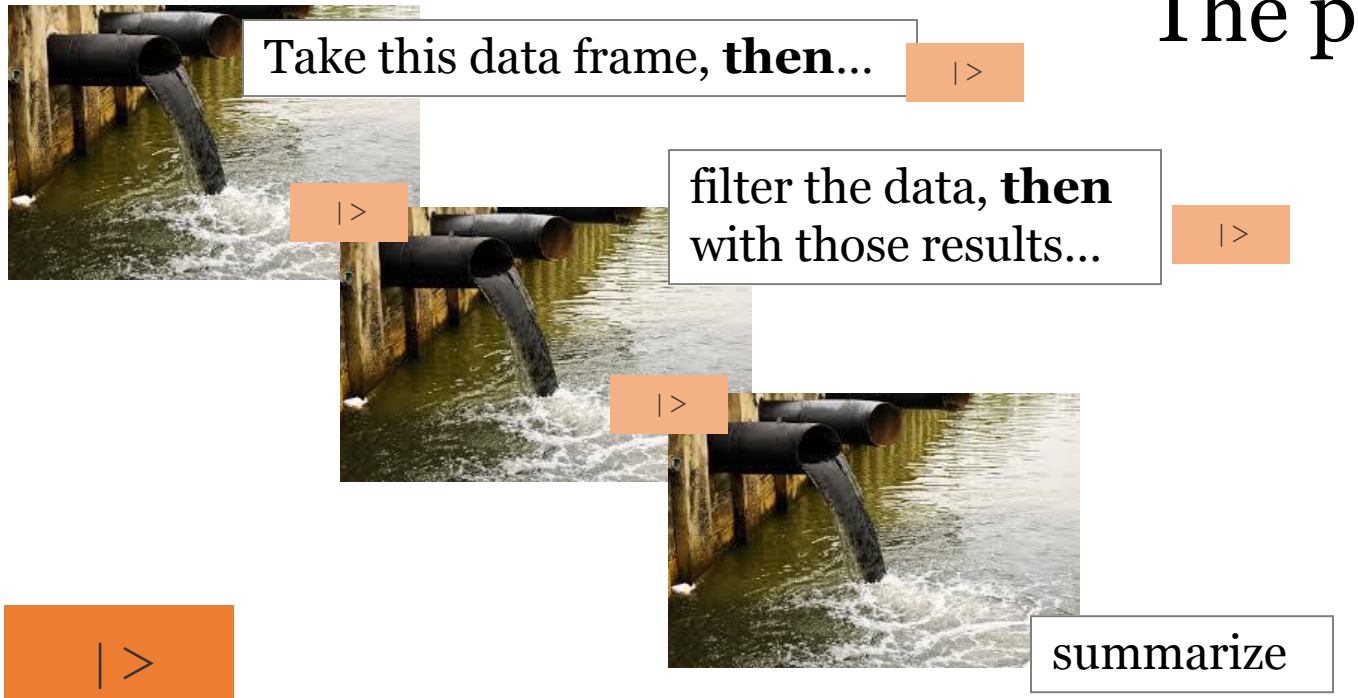
Data wrangling verbs

The native pipe `|>` *and the verbs in the* ***dplyr*** *package*

filter()
summarize()
group_by()
mutate()
arrange()
select()

The pipe operator



|>

|>

- “Pipe” a data frame into a “verb” command
- “Chain” the results from one “verb” command into another
- Think of it as the word **“then”**

filter()

Pull out just the **rows** you want in a data set, based on some criteria

Subset Observations (Rows)



filter()

Subset Observations (Rows)



Example

only keep rows of the data in which the column meets this criteria

the data frame

```
on_time_flights <- flights |>
  filter(arr_delay < 30)
```

the verb (function)

what column (variable) to use for filtering

filter()

Comparisons: `==` `!=` `<` `>` `<=` `>=`, combined with `&` (and) or `|` (or)



Another example

only keep rows of the data in which the column meets this criteria

the data frame

```
newark_flights <- flights |>
  filter(origin == "EWR")
```

the verb (function)

what column (variable) to use for filtering

summarize()

Take a column of data from a data frame and reduce it down to a single summary statistic



summarize()



Example

the data frame

what summary statistic to calculate

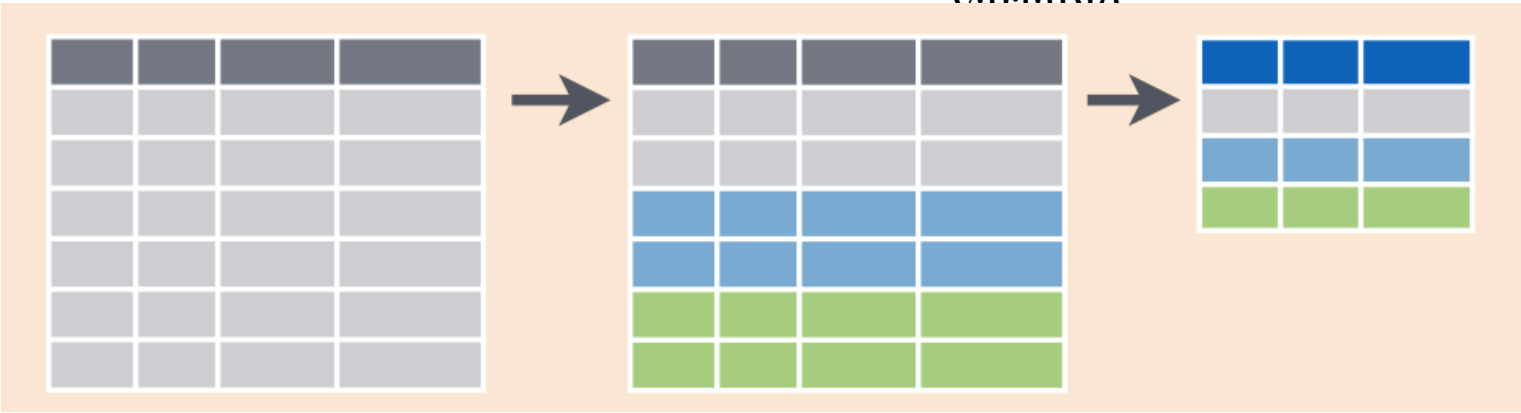
```
flights |>
  summarize(av_dist = mean(distance))
```

the verb (function) what to call the result

the column to summarize

group_by() then summarize()

Take a column of data and reduce it down to a summary statistic, by some grouping variable



group_by() then su



Example

the data frame what to group by

```
flights |>
  group_by(origin) |>
  summarize(av_dist = mean(distance))
```

the verb (function) what to call the result

the column to summarize

mutate()

Add a new column to a data frame, computed from the columns already there



mutate()

Make New Variables



Example

the data frame

```
flights |>
  mutate(dist_km = distance * 1.609)
```

the verb (function)

what to call the new column (new variable)

what to put in the new column